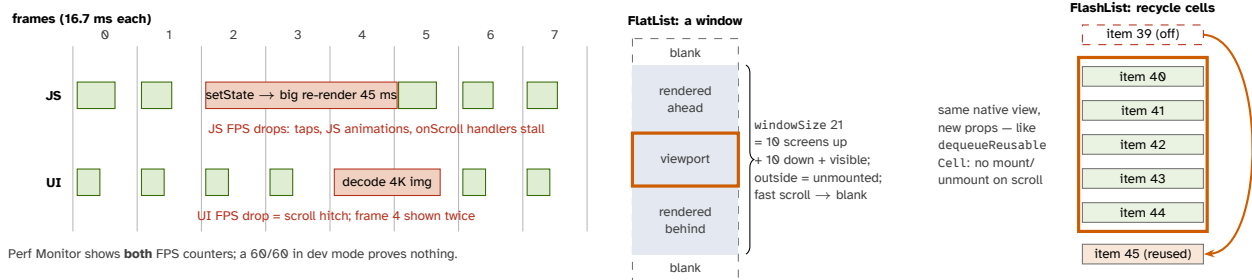


**In one line:** An RN app drops frames on **two** independent threads: the **JS thread** (React render, business logic, JS-driven animation) and the **UI/main thread** (native layout commit, drawing, native animation). Each gets **16.7 ms per frame at 60 Hz** (8.3 ms at 120 Hz ProMotion). Find *which* thread is late, *prove* it with a profiler on a **release** build, then apply the cheapest fix.



## What drops frames on each thread

- JS:** re-rendering big subtrees, heavy parse/sort/filter in render, Context/store fan-out, `console.log` in loops, JS-driven animations.
- UI:** deep/overdrawn trees, huge image decodes, many views mounting at once, sync native-module work on main — diagnose as in UIKit: *Instruments*.
- New Arch (Fabric + JSI) removes the async JSON bridge, **not** the budgets: a 45 ms render is still 3 dropped JS frames.

## Lists — FlatList knobs (VirtualizedList)

- `initialNumToRender` (10) — first paint: just fill one screen.
- `maxToRenderPerBatch` (10) — fill rate vs responsiveness.
- `windowSize` (21, in viewports) — lower = less memory, more blank. `removeClippedSubviews` — detach off-screen views (Android default).
- `getItemLayout` — fixed height → no measuring, instant `scrollToIndex`. `keyExtractor` — stable ids, never the index.
- Stable renderItem** (`useCallback`) + memo row + primitive props, or every row re-renders on every parent render.
- FlashList** (Shopify) recycles cells; **v2:** New Architecture only, JS-only, **no size estimates** (v1 needed `estimatedItemSize`); `getItemType` = separate pools per row kind. Trap: `useState` in a recycled cell leaks into the next item.

## Example — a list row that does not re-render

```
const Row = memo(({ id, title, onPress }: P) => (
  <Pressable onPress={() => onPress(id)}>
    <Text>{title}</Text></Pressable>);
function Feed({ items }: { items: Item[] }) {
  const onPress = useCallback((id: string) => open(id, []));
  const renderItem = useCallback(({ item }) => ( // stable
    <Row id={item.id} title={item.title}
      onPress={onPress} />), [onPress]);
  return <FlatList data={items} renderItem={renderItem}
    keyExtractor={(i) => i.id} // not index
    getItemLayout={(_, i) =>
      ({ length: 64, offset: 64 * i, index: i })}
    windowSize={7} />;
}
```

## Profiling toolbox

React Native DevTools (Profiler, Memory/heap snapshots; default since 0.76) · Hermes **sampling profiler** · Perf Monitor · Xcode *Time Profiler*, *Animation Hitches*, *Allocations/Leaks* · Android Studio profilers, Peretto. Test on a low-end Android.

## Re-renders, memoisation, heavy work

- Find: React DevTools **Profiler** (flame chart, “why did this render?”, highlight updates), `why-did-you-render`.
- Memoise at **boundaries** (rows, heavy subtrees): memo is useless if the parent passes a new object/closure each render. React Compiler automates it.
- Context:** every consumer re-renders on a new value — split by change rate, `useMemo` the value, or use a selector store.
- Off JS: native/JSI (C++), worklets, the server; `startTransition`; chunk long loops; defer until after the navigation transition.

## Images · startup · bundle · leaks

- Images:** decoded size =  $w \times h \times 4$  bytes — ask the CDN for display size. `expo-image` (SDWebImage/Glide, memory+disk cache, `cachePolicy`) or `FastImage`.
- Startup:** Hermes ships **bytecode** (no parse at launch, `mmap'd`); Metro `inlineRequires` + `lazy require()` defer module init; TurboModules load on first use (legacy modules init eagerly).
- Bundle:** source-map visualiser; drop giant deps, whole-lodash imports.
- Leaks:** listeners without `sub.remove()` in effect cleanup, uncleared `setInterval`, closures pinning big data, mounted screen stacks.

## Traps

- Measuring in dev mode:** `__DEV__` checks, no minify, debugger attached — often several × slower. Profile a **release** build.
- “`useMemo` everywhere” — memo costs; fix the unstable prop. `key={index}` — reorders reuse the wrong row state.
- Native driver/Reanimated keep animation smooth while JS is busy, but the tap handler is *still* late — the JS problem remains.

## Remember

**Which thread? Release build? Measure, then fix the cause:**  
fewer renders · fewer views · smaller images · less JS at startup.